

Experiment 34

Aim

To perform morphological Top Hat operation on an image using OpenCV.

Procedure

1. Import the required libraries, OpenCV and NumPy.
2. Read the input image.
3. Convert the image into grayscale.
4. Create a morphological kernel.
5. Apply the Top Hat operation using `cv2.morphologyEx()`.
6. Display the original and Top Hat images.
7. Save the resulting image.

Code:

```
import cv2

import numpy as np

img = cv2.imread("image.jpg")

if img is None:

    print("Error: Image not found!")

    exit()

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

kernel = np.ones((5, 5), np.uint8)

top_hat = cv2.morphologyEx(gray, cv2.MORPH_TOPHAT, kernel)

cv2.imshow("Original Image", img)

cv2.imshow("Top Hat Image", top_hat)
```

```
cv2.imwrite("top_hat_image.jpg", top_hat)
```

```
cv2.waitKey(0)
```

```
cv2.destroyAllWindows()
```

Sample Input:



Output:



Result

The morphological Top Hat operation was successfully performed using OpenCV. The Top Hat operation highlighted small bright regions and details in the image.